

Seat No:

Enrollment No:

PARUL UNIVERSITY
FACULTY OF ENGINEERING & TECHNOLOGY
B.Tech, Winter 2025-26 Examination

Semester: V

Subject Code: 303105301

Subject Name: AWS Fundamental

Date: 19-11-2025

Time: 2.00 pm to 4.30 pm

Total Marks: 60

Instructions:

1. This question paper comprises of two sections. Write answer of both the sections in separate answer books.
2. From Section A, **Q.1** is compulsory, From Section B, **Q.1** is compulsory.
3. Figures to the right indicate full marks
4. Draw neat and clean drawings & Make suitable assumptions wherever necessary.
5. Start new question on new page.
6. BT- Blooms Taxonomy Levels – Remember-1, Understand -2, Apply-3, Analyse-4, Evaluate-5, Create-6

SECTION - A

Q.1	Answer the following questions.	Marks	CO	BT
	A. a) What is Lambda? b) Define Serverless Computing. c) State one use case of Lambda.	06	CO1	BT1
	B. a) Mention two characteristics of AWS Lambda. b) Give one benefit of using Lambda. c) State one limitation of Lambda.	06	CO3	BT2
Q.2	A. Define Cloud Computing. Explain its key characteristics and give two real-world examples.	04	CO1	BT4
	B. Explain the concept of Infrastructure as a Service (IaaS) and discuss two benefits scalability and cost efficiency in brief.	05	CO1	BT1
OR				
	B. Explain the concept of Amazon VPC and discuss at least two key benefits it provides for deploying secure cloud resources.	05	CO2	BT1
Q.3	A. Explain the process of deploying a sample application to AWS Elastic Beanstalk, highlighting one key configuration required.	04	CO3	BT2
	B. Define Infrastructure as a Service (IaaS) and differentiate it from Platform as a Service (PaaS) and Software as a Service (SaaS).	05	CO3	BT4
OR				
	B. Harsha is tasked with designing a scalable and highly available web application architecture. Tasks: Create an Auto Scaling group with minimum 1 and maximum 4 EC2 instances using a launch template (Amazon Linux 2023, t2.micro). Configure an Application Load Balancer (ALB) listener on port 80. Attach the Auto Scaling group to the ALB target group. Simulate traffic load and verify automatic scaling and proper traffic distribution.	05	CO3	BT1

SECTION - B

Q.1	Answer the following questions.	Marks	CO	BT
	A. a) Define a computer network. b) Name two types of networks. c) Give one real-life example of networking.	06	CO2	BT2

	B. (a) What is Amazon S3? (b) Name any two storage classes of Amazon S3. (c) State one real-world use case of Amazon S3.	06	CO4	BT1
Q.2	A. Propose an optimized AWS backup strategy for a financial institution and justify your approach.	04	CO4	BT2
	B. Implement IAM roles for granting EC2 access to Amazon S3. Tasks: • Create an IAM role with AmazonS3ReadOnlyAccess. • Attach the role to an EC2 instance. • Verify access to an S3 bucket using AWS CLI commands.	05	CO2	BT3
	OR			
	B. Deploy Python Lambda + S3 trigger for data processing. Tasks: • Create S3 bucket incoming-data. • Create Lambda function to process uploaded files. • Configure S3 Event to trigger Lambda. • Upload sample file and verify output.	05	CO4	BT5
Q.3	A. Compare IAM users vs IAM roles in terms of best practices.	04	CO5	BT6
	B. Describe the Private Cloud model and explain scenarios where it is preferable, highlighting one key advantage.	05	CO1	BT3
	OR			
	B. Explain the step-by-step process to launch an EC2 instance.	05	CO3	BT3